Four universally separated marks in Tait-colourable cubic graphs

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AI-assistance disclosure. OpenAI Codex agents (GPT-5-series models, accessed 26 July 2026) formulated the marked theorem, found the proof architecture, supplied the cut and decomposition arguments, checked the cited source statements, and drafted this manuscript under human direction. An additional Codex agent performed the publication audit, but that is not independent human verification or peer review. No human author or referee is represented as having certified the theorem, proof, or novelty search. This is a research draft, not a submission-ready paper. A human author must check every reduction and source, conduct an independent literature review, and accept ordinary scholarly responsibility before public submission.

Abstract

Let H be a connected simple cubic graph admitting a proper three-edge-colouring, and let S be a matching of four edges. Suppose that in every proper three-edge-colouring, no bichromatic cycle contains two members of S. Suppose also that every vertex set X for which $H[X]$ contains a cycle satisfies

$$|\delta_H(X)| + |S \cap E(H[X])| \geq 4.$$

We prove that H has an even edge-subgraph containing S such that every cycle component contains an even number of members of S. More precisely, the proof produces either one cycle through all four marks or two disjoint cycles containing two marks each.

The proof first resolves cyclic cuts of size two and all marked cyclic cuts of size three. In the remaining case, a self-contained cubic three-cut decomposition yields cyclically 4-edge-connected atoms. A central atom is selected by the distribution of the four marks. Kempe switching gives a Tait colouring in which all marks have one colour. The prescribed-edge theorems of Aldred, Ellingham, Hemminger, and Holton and of Knappe and Pitz supply the central cycle, according to whether the four selected edges remain distinct. One-mark paths lift it through the branches.

The argument is computer-free. It proves the stated marked-graph theorem only; it neither proves nor refutes the five-cycle double cover conjecture. The exact novelty claim is provisional pending independent expert review.

1 Introduction

A *Tait colouring* of a cubic graph is a proper edge-colouring with three colours. For a fixed Tait colouring, each pair of colours induces a disjoint union of cycles. We call a matching S *universally separated* when, in every Tait colouring, each such bichromatic cycle contains at most one edge of S.

The theorem below arose in a reduction related to five-cycle double covers. That motivation is not used in the proof. In particular, the standard five-cycle double cover conjecture remains a separate open problem; see Oum's formulation as Conjecture 18 [4, Section 9.1].

Theorem 1.1 (four-mark theorem)**.** *Let H be a connected simple Tait-colourable cubic graph, and let $S \subseteq E(H)$ be a universally separated matching of size four. Assume that*

$$|\delta_H(X)| + |S \cap E(H[X])| \geq 4 \tag{M}$$

whenever $H[X]$ contains a cycle. Then H has a binary cycle $Q \subseteq E(H)$ such that

1. *$S \subseteq Q$; and*
2. *every component of Q contains an even number of edges of S.*

More precisely, Q can be chosen either as one cycle containing all four marks or as the union of two disjoint cycles, each containing two marks.

Here a *binary cycle* means an edge-subset of even degree at every vertex. In a cubic graph its nonempty components are cycles.

The two external cycle results used below are exact and classical. Knappe and Pitz prove, in particular, that in a 3-edge-connected graph a set of at most three edges lies in a circuit if and only if it contains no odd edge cut [2, Fact 5.4]. Aldred, Ellingham, Hemminger, and Holton prove that four independent edges of a quasi 4-connected graph lie on a common cycle [1, Theorem 3.3]. They also record that cyclically 4-edge-connected cubic graphs are quasi 4-connected.

Related general tree decompositions of 3-connected graphs into cyclically 4-edge-connected components were developed by Pastor [5]; the exact cubic construction also appears in Nedela, Seifrtová, and Škoviera [3, Section 6]. To avoid any convention or hypothesis gap, we prove the elementary cubic specialization needed here.

We found no prior statement of Theorem 1.1 in the sources above or in targeted searches using its marked-colouring and cut hypotheses. This is only a provisional novelty assessment. The prescribed-edge cycle theorems, Kempe switching, cut parity, and three-cut decomposition are not claimed as new.

2 Colour flexibility and pole lemmas

Lemma 2.1 (mark precolouring)**.** *Let S be universally separated in a Tait-colourable cubic graph. Every map from S to the three colours extends to a Tait colouring.*

Proof. Start with any Tait colouring. Suppose that a mark e currently has colour p and is prescribed colour $q \neq p$. The unique pq-bichromatic cycle through e contains no other mark. Interchanging p and q on that cycle changes the colour of e without changing another member of S. Process the marks one at a time. Universal separation applies after every switch, so a later switch does not disturb a mark already processed. □

Lemma 2.2 (cut and cap facts)**.** *Let G be a simple Tait-colourable cubic graph.*

1. *G is bridgeless.*
2. *In a Tait colouring, the three edges of a three-edge cut have three distinct colours.*
3. *If G has no edge cut of size at most two and B is a cyclic three-edge cut, then the three edges of B have distinct ends on each shore.*
4. *Capping either shore of B by one new vertex adjacent to its three boundary ends produces a simple cubic* 3*-edge-connected graph.*

Proof. Every edge lies on either of the two bichromatic cycles using its colour, which proves (1). For (2), if X is a shore, then the number of edges of each colour in $\delta_G(X)$ has the same parity as $|X|$. A cubic shore of a three-edge cut has odd order, so each colour occurs oddly; all three therefore occur once.

For (3), suppose two cut edges have a common end v on a shore A. The sole edge from v into $A - v$, together with the third cut edge, separates $A - v$. Every cycle of $G[A]$ avoids v, since v has only one neighbour in A. This gives a cyclic cut of size at most two, a contradiction.

The cap is simple by (3). If a set Y defines a cut of size at most two in the cap, choose the shore not containing the new cap vertex. Replacing cap edges by the corresponding members of B gives an edge cut of the same size in G. This proves (4). □

We use a *k-pole* only as shorthand for a connected induced shore of a k-edge cut, with the cut edges regarded as boundary semiedges. The boundary ends below are distinct.

Lemma 2.3 (two-mark shore)**.** *Let $P = H[A]$ be a connected two- or three-pole of minimum degree two. Suppose exactly two marks lie internally in P, and (M) holds for every cyclic vertex set contained in A. Then one cycle of P contains both marks.*

Proof. Delete the bridges of P, and form the tree of bridge-components. If P has no bridge, then it is 2-edge-connected. A 2-edge-connected subcubic graph of minimum degree two has no cutvertex: two components behind a cutvertex would each require at least two incident edges at that vertex. Thus P is 2-connected. Subdivide the two marks. Two vertices of a 2-connected graph lie on a common cycle, which gives the required cycle after undoing the subdivisions.

Now suppose the bridge-component tree is nontrivial. Every leaf component K is cyclic. It cannot be an isolated vertex, since it has one incident bridge but every vertex of P has degree at least two. Let $p(K)$ count pole boundary edges incident with K, and let $s(K)$ count its internal marks. Exactly one bridge leaves K, so (M) gives

$$1 + p(K) + s(K) \geq 4.$$

If K does not contain both marks, then $s(K) \leq 1$, and hence $p(K) \geq 2$. Two such leaves would consume at least four boundary edges, impossible for a two- or three-pole. Some leaf component therefore contains both marks. It is 2-connected by the preceding argument and has a cycle through them. □

Lemma 2.4 (one-mark shore)**.** *Let $P = H[A]$ be a connected three-pole of minimum degree two with boundary ends u_1, u_2, u_3. Suppose exactly one mark e lies internally in P, and (M) holds for cyclic vertex sets contained in A. For every pair $i \neq j$, P has a u_i-to-u_j path containing e.*

Proof. The pole has no bridge. Otherwise, a leaf K of its bridge-component tree is cyclic as in Lemma 2.3. The inequality

$$1 + p(K) + s(K) \geq 4$$

says that an unmarked leaf consumes all three boundary edges and a marked leaf consumes at least two. Any two leaves therefore consume more than the three available boundary edges.

Thus P is 2-edge-connected and hence 2-connected by the subcubic cutvertex argument. Add a new edge u_iu_j, allowing a parallel edge temporarily if necessary. In a 2-connected graph, any two edges lie on a common cycle. A cycle through the new edge and e, with the new edge removed, is the required path. □

3 Low cyclic cuts

Lemma 3.1 (cyclic two-cuts)**.** *If H satisfies the hypotheses of Theorem 1.1 and has a cyclic two-edge cut, then the conclusion of the theorem holds.*

Proof. Apply (M) to both shores. Each shore contains at least two internal marks. Since there are only four marks, the cut is unmarked and each shore contains exactly two. Lemma 2.3 gives a cycle through the two marks on each shore. Their disjoint union is the required binary cycle. □

Lemma 3.2 (cyclic three-cuts)**.** *Let H satisfy the hypotheses of Theorem 1.1, have no cyclic two-edge cut, and let B be a cyclic three-edge cut. Unless B is unmarked and splits the internal marks $1+3$, the conclusion of Theorem 1.1 holds.*

Proof. Let $r = |B \cap S|$, and let s_A, s_D count internal marks on the two shores. Inequality (M) gives $s_A, s_D \geq 1$, while

$$r + s_A + s_D = 4.$$

Hence $r \leq 2$.

If $r = 2$, each shore contains one internal mark. The two marked members of B determine the same boundary pair on both shores. Lemma 2.4 supplies a path through the internal mark on each side for that pair. The paths and the two marked cut edges glue to a cycle through all four marks.

Suppose $r = 1$, with two internal marks on A and one on D. Cap A by a new vertex z. The cap is simple and 3-edge-connected by Lemma 2.2. Let F consist of the cap edge corresponding to the marked member of B, together with the two internal marks of A. These are three independent edges.

The set F contains no odd edge cut of the cap. Indeed, edge connectivity leaves only the possibility that F itself is a three-edge cut. Since F is independent, this cut is nontrivial and both shores are cyclic: a connected shore of order $n > 1$ has $(n-1)/2 \geq 1$ independent cycles by the cubic degree sum. Choose its shore X not containing z. Viewed in H, it has $|\delta_H(X)| = 3$, while both internal marks of A cross the cut and no mark lies internally in X. This contradicts (M).

Knappe and Pitz's $g(3) = 3$ result [2, Fact 5.4] now gives a circuit through F. Because the cap is cubic, the circuit is a cycle. At z it uses the marked cap edge and one other cap edge. Deleting z leaves a path through the two internal marks for that boundary pair. Lemma 2.4 supplies a path through the sole internal mark of D for the same pair. Gluing the paths through the corresponding two members of B gives a cycle containing all four marks.

It remains that $r = 0$. The internal split is $1+3$ or $2+2$. In the $2+2$ case, Lemma 2.3 on both shores gives two closed cycles whose union proves the theorem. The only unresolved local case is therefore an unmarked $1+3$ cut. □

4 A cubic atom tree

For a cyclic three-edge cut, *capping a shore* means adjoining one new cubic vertex at its three boundary ends. Conversely, two capped graphs with specified cap vertices can be joined by deleting the cap vertices and matching their three incident ports. We call this a *three-sum*. The matching of ports is retained as part of the data.

Lemma 4.1 (cubic atom tree)**.** *Let G be a simple cubic graph with no edge cut of size at most two. There is a finite tree T with a simple cubic graph G_t at every node t such that:*

1. *each G_t is 3-edge-connected and has no cyclic three-edge cut;*

2. *G is recovered by iterated three-sums along the edges of T; and*

3. *each edge of T corresponds, after expanding all other factors, to a cyclic three-edge cut of G.*

Every G_t is a cyclically 4-edge-connected cubic graph and hence is quasi 4-connected.

Proof. Proceed recursively. If G has no cyclic three-edge cut, use the one-node tree. Otherwise choose such a cut, cap its two shores, and recurse on the two capped graphs. Lemma 2.2 shows that both are simple, cubic, and 3-edge-connected. Each has fewer vertices than G, so the recursion terminates.

In each recursive tree, the cap vertex introduced by the chosen cut belongs to one terminal factor. Join those two factor nodes by a new tree edge and remember the port matching. Reversing this operation recovers G, proving (2).

For (3), expand the caps on either side of a recursive cut. Put an expanded branch on the same side as its cap vertex. Every cap edge in the cut is replaced by exactly one original edge, and all other port edges remain internal. Thus the boundary still has size three. A cycle using a cap vertex lifts through a path in the connected expanded branch between the two used ports; a cycle avoiding the cap is unchanged. Cyclicity of both shores is therefore preserved.

It remains only to justify the last sentence. A simple 3-edge-connected cubic graph with no cyclic three-edge cut is 3-vertex-connected. A cutvertex would force two components each to use at least two of its three incident edges. If $\{u, v\}$ were a two-vertex cut, then u and v could not be adjacent; the six edges from them would split into two three-edge boundaries. Simplicity forces both corresponding shores to contain cycles, contrary to the absence of a cyclic three-edge cut. The factor is consequently cyclically 4-edge-connected in the standard cubic sense. Aldred et al. observe that every such cubic graph is quasi 4-connected [1]. □

Lemma 4.2 (central node)**.** *Let nonnegative integer weights of total four be assigned to the vertices of a finite tree T. Suppose every tree edge separates the total weight as $1+3$. Then T has a vertex a such that every component of $T-a$ has weight one. If a has weight r and degree q, then $r+q=4$.*

Proof. Orient every edge toward its side of weight three. A finite directed tree has a sink a. The side away from a across each incident edge has weight one. These are exactly the components of $T-a$, so their number is q and $r+q=4$. □

5 Proof of the four-mark theorem

Proof of Theorem 1.1. Suppose for a contradiction that the required binary cycle does not exist. Lemma 3.1 eliminates cyclic two-edge cuts. A Tait-colourable cubic graph is bridgeless, and every two-edge cut in a simple bridgeless cubic graph is cyclic. Indeed, a minimal two-edge cut has connected shores, and a shore of order n has

$$|E(H[X])| - |X| + 1 = n/2 > 0$$

by the cubic degree sum; the same holds on the other shore. Hence H has no edge cut of size at most two.

Apply Lemma 4.1. Every tree edge expands to a cyclic three-edge cut of H. By Lemma 3.2, each such cut is unmarked and splits the four marks $1+3$. Since no decomposing cut is marked, every mark belongs internally to a unique atom. Weight each atom by its number of marks. Lemma 4.2 gives a central atom A, containing r internal marks and incident with $q = 4 - r$ branches, each containing exactly one mark.

By Lemma 2.1, choose a Tait colouring of H in which all four marks have one colour, say c. The three edges of every decomposing cut have distinct colours. The colouring therefore induces a Tait colouring on every capped atom by retaining those colours on the three cap edges.

In A, select its r internal marked edges and, at each of its q cap vertices, select the unique incident edge of colour c. Let F be the set of distinct selected edges. Because all members of F have colour c, F is a matching; it contains every internal mark, meets every cap vertex, and has size at most four. Two cap vertices can be adjacent. In that case their unique incident c-edge may be the same edge, which is why F is defined as a set of distinct edges and may have size less than four. A cycle containing that shared edge nevertheless visits both cap vertices.

There is a cycle C of A containing F. If $|F| = 4$, this is the independent-edge theorem of Aldred, Ellingham, Hemminger, and Holton for quasi 4-connected graphs [1, Theorem 3.3]. If $|F| \leq 3$, then F contains no odd cut: a one-edge cut is excluded by 3-edge-connectivity, and a three-edge cut contained in a matching would be nontrivial and cyclic, which does not occur in an atom. Knappe and Pitz's $g(3) = 3$ theorem [2, Fact 5.4] gives a circuit through F, and in the cubic graph A its edge set is a cycle.

The cycle C visits every cap vertex, since it contains a selected edge incident with that vertex. At a cap vertex it uses exactly two ports. Expand the corresponding component of the atom tree. In the original graph this component is a connected three-pole containing exactly one mark. It satisfies (M) on every cyclic subset, so Lemma 2.4 gives a path through its mark between the two ports used by C. Replace the two-edge passage through the cap vertex by that path and the corresponding boundary edges.

Perform this replacement for all q branches. Formally, first reconstruct each component of $T - A$ internally while retaining its root cap, and then attach those branch composites to A one at a time. If two cap vertices of A are adjacent, expanding the first deletes their shared cycle edge and reconnects its dangling end to the inserted branch. The resulting edge is still the same port of the lifted cycle at the second cap vertex; that vertex remains cubic and on the cycle. Expanding the second cap later replaces its two current ports in exactly the same way. Thus adjacency of cap vertices causes no ambiguity and no splitting of the cycle.

Each sequential replacement of a cap passage by the corresponding one-mark path preserves a single cycle. The final cycle lies in H and contains the r central marks and the one mark from each of the q branches, hence all four members of S. This contradicts the assumed failure and proves the theorem. □

Remark 5.1 (scope of the one-cycle conclusion). The central-atom argument and the marked three-cut cases construct one cycle through all four marks. The proof may instead terminate at an unmarked cyclic three-cut with a $2 + 2$ mark split, where Lemma 3.2 produces a binary cycle with two components. Thus this manuscript does *not* claim the stronger statement that one cycle always contains all four marks. The precise alternative proved is one four-mark cycle or two disjoint two-mark cycles.

AI-use statement

The page-one disclosure is part of this draft and should remain in every derivative version. Codex agents materially contributed to the theorem statement, proof search, low-cut lemmas, atom-tree proof, source audit, and exposition. The proof is written for direct human checking and uses no finite computation, SAT solver, or unverifiable certificate. That makes it human-checkable; it does not mean a human has checked it. No independent human verification is asserted.

The precise source-dependent inputs are the prescribed-circuit result of Knappe and Pitz [2], the quasi-4-connected prescribed-cycle result of Aldred, Ellingham, Hemminger, and Holton [1], and, for context only, the broader decomposition theory [5, 3]. Any prospective human author should verify those applications against the published papers and repeat the novelty search before submission.

References

[1] R. E. L. Aldred, M. N. Ellingham, R. L. Hemminger, and D. A. Holton. Cycles in quasi 4-connected graphs. *Australasian Journal of Combinatorics*, 15:37–46, 1997.

[2] Paul Knappe and Max Pitz. Circuits through prescribed edges. *Journal of Graph Theory*, 93(4):470–482, 2020.

[3] Roman Nedela, Michaela Seifrtová, and Martin Škoviera. Decycling cubic graphs. *Discrete Mathematics*, 347(8):114039, 2024.

[4] Sang-il Oum. A proof of the cycle double cover conjecture by OpenAI: An exposition, 2026. Version 2, revised 24 July 2026.

[5] A. V. Pastor. On a decomposition of a 3-connected graph into cyclically 4-edge-connected components. *Journal of Mathematical Sciences*, 232(1):61–83, 2018.